

EdgeCount: Statistical Theory and Algorithmic Efficiency

Joel R. Pradines

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Abstract

This vignette details the statistical framework underlying the **EdgeCount** package. First, the Random Graph with Given Expected Degrees (RGGED) model is described. Second, the derivation of edge-count statistics is presented. Third, the algebraic optimizations that allow the package to calculate connectedness statistics with linear time complexity are explained. Last, the implementation of vectorized functions contained in the package is discussed, and it is shown that they have optimal time complexity.

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1 Introduction

The **EdgeCount** package provides methods to analyze the connectedness of vertex sets in large, sparse, undirected graphs. Whether analyzing interaction networks or bipartite group-element graphs, the question is:

Is an observed number of edges, called edge count, within a set (or between two sets) significantly higher than expected by chance?

To answer this, two components must be implemented.

First, the random experiment defining the concept of *chance* must be clearly specified and, in general, simulated to provide a control distribution.

Second, one needs to be able to fairly compare edge-count values to each other. A fair comparison requires accounting for vertex degrees, as these may vary by orders of magnitude. This is achieved here by using a null model of random graphs that preserves the expected degrees of vertices [3, 4]. Specifically, the average degree of a vertex over all random graph realizations matches its observed value in the original network. The analytical expressions for the likelihoods of edge-count variables in this model were derived in [13]. These expressions enable fair and fast comparisons of edge-count values.

2 Random graph model

Let $G(V, E)$ be an undirected graph with no loops and no multiple edges.

- Let $N = |V|$ be the order of the graph (number of vertices).

- Let $M = |E|$ be the size of the graph (number of edges).
- Let k_i be the degree of vertex v_i .
- Let $\langle k \rangle$ be the the average vertex degree.

Call degree sequence of G the nonincreasing sequence of all vertex degrees. An obvious null model for G is that of a random graph whose realizations always preserve the degree sequence. This model is referred to as the Random Graph with Fixed Degree Sequence (RGFDS). However, it is known that the analytical treatment of the RGFDS is a difficult problem [2, 11, 5, 12]. Namely, deriving simple analytical expressions for the likelihoods of edge-count variables is in general hard with the RGFDS. This implies that slow numerical methods based on simulations must be used [8, 10, 9].

A less constrained random graph model than the RGFDS is the Random Graph with Given Expected Degrees (RGGED) [3, 4]. While an individual realization of the RGGED might not preserve a vertex degree, it is preserved on average over all realizations. The key feature of the RGGED is that edges are treated as independent random variables, enabling the derivation of analytical expressions for the distributions of edge-count variables [13].

In the RGGED model, edges are treated as independent Bernoulli random variables: an edge $v_i v_j$ exists with probability p_{ij} (Bernoulli parameter) and does not exist with probability $1 - p_{ij}$. When the graph is large and sparse on average over its vertices ($\langle k \rangle^2 \ll 2M$), the probability p_{ij} of observing an edge between vertices v_i and v_j is given by [13]:

$$p_{ij} \simeq 1 - \exp(-k_i k_j / (2M)). \quad (1)$$

Useful upper bounds for p_{ij} are

$$p_{ij} \leq \min\left(1, \frac{k_i k_j}{2M}\right) \leq \frac{k_i k_j}{2M}. \quad (2)$$

2.1 Within and between variables

The `EdgeCount` package considers two types of edge-count variables:

- "within" for the number z of edges observed in a given set S of vertices
- "between" for the number z of edges observed between two sets S_1 and S_2 of vertices

Observed values z are obviously dependent on set sizes, but they should also be made conditional to degrees of the vertices in the set(s). Hence, to an observed value z is associated a corresponding random variable Z in the RGGED.

2.2 Edge-count distribution

Edge-count random variables in the RGGED model of a large and sparse graph have approximate Poisson distribution. This is because they are the sum of independent Bernoulli random variables with small parameters $p_{ij} \ll 1$, even when these parameters have different values [7, 6, 13].

Call Z an edge-count random variable defined by the sum of m potential edges, each edge corresponding to a pair $v_i v_j$ of vertices:

$$Z = \sum_{u=1}^m B_u \quad \text{with} \quad B_u \sim \text{Bernoulli}(p_u = p_{ij}). \quad (3)$$

Then, we have $Z \sim \text{Poisson}(\lambda, m)$ as defined by

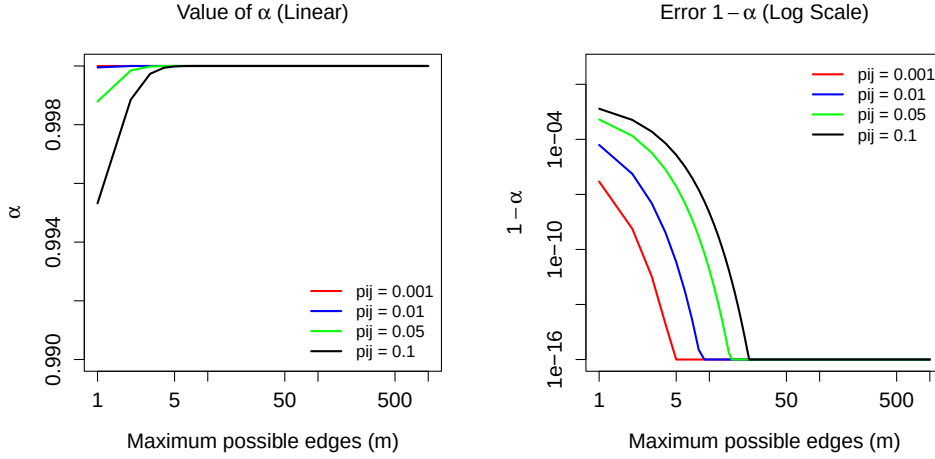
$$\Pr(Z = z) = \frac{e^{-\lambda}}{\alpha} \frac{\lambda^z}{z!} \quad \text{with} \quad \lambda = \sum_{u=1}^m p_u \quad \text{and} \quad \alpha = e^{-\lambda} \sum_{x=0}^m \frac{\lambda^x}{x!}. \quad (4)$$

The normalization factor α is utilized because an observed number z of edges is bounded by a maximum number m of edges. This said, we have $\alpha \simeq 1$ when $p_{ij} \ll 1$. This is numerically illustrated next.

```

> # Function to calculate alpha and its complement
> get_alpha_stats <- function(m, pij_avg) {
+   lambda <- m * pij_avg
+   alpha <- stats::ppois(m, lambda)
+   alpha_comp <- stats::ppois(m, lambda, lower.tail = FALSE)
+   return(c(alpha, alpha_comp))
+ }
> # Parameters
> pij_values <- c(0.001, 0.01, 0.05, 0.1)
> m_values <- c(1:29, seq(30, 1000, by = 10))
> colors <- c("red", "blue", "green", "black")
> # Plots
> par(mfrow=c(1, 2), mar=c(5, 5, 4, 2) + 0.1, cex.lab=1.25,
+     cex.axis=1.25, cex.main=1.25)
> plot(1, type="n", xlab="Maximum possible edges (m)",
+      ylab=expression(alpha),
+      xlim=c(1, 1000), ylim=c(0.99, 1.00),
+      log="x",
+      main=expression(paste("Value of ", alpha, " (Linear)")))
> for(i in seq_along(pij_values)) {
+   lambdas <- m_values * pij_values[i]
+   y_vals <- ppois(m_values, lambdas)
+   lines(m_values, y_vals, col=colors[i], lwd=2)
+ }
> legend("bottomright", legend=paste("pij =", pij_values),
+       col=colors, lwd=2, bty="n", cex=1)
> epsilon <- 1e-16
> plot(1, type="n", xlab="Maximum possible edges (m)",
+      ylab=expression(1 - alpha),
+      xlim=c(1, 1000), ylim=c(epsilon, 1),
+      log="xy",
+      main=expression(paste("Error ", 1 - alpha, " (Log Scale)")))
> for(i in seq_along(pij_values)) {
+   lambdas <- m_values * pij_values[i]
+   y_vals <- ppois(m_values, lambdas, lower.tail = FALSE)
+   y_vals <- pmax(y_vals, epsilon)
+   lines(m_values, y_vals, col=colors[i], lwd=2)
+ }
> legend("topright", legend=paste("pij =", pij_values),
+       col=colors, lwd=2, bty="n", cex=1)

```



For a within variable associated with a set S we have

$$m = |S|(|S| - 1)/2 \quad \text{and} \quad \lambda = \sum_{i < j, i, j \in S} p_{ij}. \quad (5)$$

For a between variable associated with two sets S_1 and S_2 we have

$$m = |S_1||S_2| \quad \text{and} \quad \lambda = \sum_{i \in S_1, j \in S_2} p_{ij}. \quad (6)$$

Since the distribution of an edge-count variable is easily computed, p-values are also calculated.

$$\Pr(Z \geq z) = \sum_{x=z}^m \Pr(Z = x). \quad (7)$$

P-values are calculated with R function `ppois`.

```
> alpha <- stats::ppois(m, lambda, lower.tail = TRUE)
> p_value <- (stats::ppois(z - 1, lambda, lower.tail = FALSE) -
+           stats::ppois(m, lambda, lower.tail = FALSE)) / alpha
```

The Poisson distribution enables quantification of the variability on vertex degree introduced by the RGGED. In the model, the degree of a vertex v_i with initial degree k_i is a random variable K_i defined by the edge count between this vertex and all the others:

$$\lambda_i = \frac{k_i}{2M} \sum_{j \neq i} k_j \simeq k_i. \quad (8)$$

That is, K_i has mean k_i and standard deviation $\sqrt{k_i}$. The introduced relative variability on k_i is reasonable: $1/\sqrt{k_i}$.

2.3 Edge-count effect size

As explained earlier, the RGGED is not directly utilized as a null model to answer a question. Rather, it is used as an intermediate step that enables fair and fast comparison of diverse observed edge counts. Such a comparison can be between two edge-count variables defined by sets of different sizes. In this case, an effect size is the most appropriate quantity for comparison.

Because edge-count random variables have Poisson distribution, the utilized effect size is

$$\log_2 \text{ Anscombe ratio: } lAr(z, \lambda) = \frac{1}{2} \log_2 \frac{z + 3/8}{\lambda + 3/8}. \quad (9)$$

The addition of $3/8$ stems from the work of Anscombe [1], who showed that $\sqrt{X + 3/8}$ is the optimal variance-stabilizing transformation for a Poisson random variable X .

3 Fast approximations for sparse graphs

A condition for sparseness in the `EdgeCount` framework is

$$\max_{i,j} \frac{k_i k_j}{2M} < 1. \quad (10)$$

This sparseness condition can be tested with method `summarize_suitability_fast`. For unipartite graphs (interaction networks), the relevant method is that of class `ECProb`. For bipartite graphs (group-element networks), the appropriate method is that of class `ECTermScoring`. The method returns a list containing the proportion `pij_over_1` of $(k_i k_j)/(2M) \geq 1$. An ideally sparse graph yields `pij_over_1` = 0.

Now, approximate p_{ij} values with $k_i k_j/(2M)$ is a conservative approach with respect to the RGGED. Namely, using this approximation

- overestimates p_{ij} values,
- overestimates edge-count p-values,
- underestimates effect sizes.

Therefore, utilizing this approximation by default is a reasonable approach.

The approximation is called **fast approximation** because it enables faster calculation of edge-count statistics.

3.1 Within set connectedness

Consider a vertex set S of size $|S| = n$. With the fast approximation the λ parameter is

$$\lambda_{in} = \sum_{i < j, i \in S} \frac{k_i k_j}{2M}. \quad (11)$$

A naive implementation would have time complexity $\mathcal{O}(n^2)$. Now, recall the expression for the square of a sum

$$\left(\sum_{i \in S} k_i \right)^2 = \sum_{i \in S} k_i^2 + 2 \sum_{i < j} k_i k_j. \quad (12)$$

Rearranging the cross terms yields

$$\sum_{i < j} k_i k_j = \frac{1}{2} \left[\left(\sum_{i \in S} k_i \right)^2 - \sum_{i \in S} k_i^2 \right]. \quad (13)$$

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